Fikisha Platform Detailed Summary Report

Business and Technical Overview

Report Date: March 16, 2026

Prepared For: Platform Owner and Operations Team

Executive Summary

Fikisha is a multi-role commerce and delivery platform with separate experiences for Customers, Merchants, Drivers, and Administrators. The platform currently runs as three coordinated applications: a main customer/staff web app, an admin operations portal, and an Express + Prisma backend API.

Core Apps	Primary Roles	Data Backbone
3	**4**	**Prisma**
Main App, Admin Portal, Backend API	Admin, Merchant, Driver, Customer	SQLite schema with role-based relations

Document purpose: consolidated summary for operations, governance, and future planning.

1. Platform Architecture

Frontend: React + TypeScript + Vite (main app) | Admin: React + MUI + Axios | Backend: Node.js + Express | ORM: Prisma | Database: SQLite | Auth: JWT + bcrypt

Application Components

Component	Purpose	Key Notes
Main App (Port 5173)	Customer storefront and role-specific staff portals	Protected routes by role, checkout and order tracking, merchant and driver dashboards
Admin Portal (Port 3000)	Operational command center	Store controls, driver fleet management, store logs and security-related views
Backend API (Port 3002)	Business logic and data access layer	Role middleware, validation, rate limiting, security headers, Prisma transactions

Data Model Coverage

Core entities include User, Store, Product, Order, OrderItem, Driver, and Category. The model supports:

- Merchant-owned stores and store-linked products
- Customer-linked order history
- Driver assignment and delivery lifecycle
- Role-based account ownership and operational boundaries

2. Role-Based Experience

Customer

- Browse stores and products
- Place orders and track statuses
- Maintain profile and local payment/address preferences

Merchant

- Access merchant-only dashboard and orders
- Manage own store products (backend-enforced)
- Store-scoped order visibility and status updates

Driver

- Driver login and task visibility
- Assigned and available-order flow
- Status progression during fulfillment

Admin

- Create and manage stores
- Open/close stores and update merchant credentials
- Manage driver fleet and credentials

3. Admin Portal Operations

Store Operations

- Create stores with merchant owner account details
- Toggle store status open/closed (without deletion as standard path)
- Edit merchant credentials securely through dedicated endpoint
- View store-level security and operational logs

Driver Fleet Management

- Add new drivers with explicit login credentials (username/password)
- Edit driver profile and credentials
- View driver order history from admin interface
- Review availability and rating summaries

Security and Audit Visibility

- Store security logs include opening/closing events and credential-change events
- Order logs include payment method, driver assignment, and timestamps
- Fallback local event storage exists in admin portal for resilience

4. Order Lifecycle and Governance

Lifecycle Stage	Status Examples	Operational Significance
Intake	PENDING, CONFIRMED	Initial acceptance and merchant acknowledgment
Preparation	PREPARING, READY_FOR_PICKUP	Store production completion and dispatch readiness
Delivery	OUT_FOR_DELIVERY	Driver handoff and last-mile movement
Closure	DELIVERED, CANCELLED	Completion or operational exception handling

> **Store Isolation:** Merchant order access is store-scoped to prevent cross-store visibility and protect commercial boundaries.

5. Security Posture

- JWT authentication with role-aware access controls
- Password hashing with bcrypt
- Rate limiting on login routes to reduce brute-force risk
- Express validator layer for input validation and schema guardrails
- Security headers via Helmet and CORS policy controls

6. Operational Readiness Summary

Area	Status	Comments
Multi-role Routing	Implemented	Customer, Merchant, Driver, and Admin route separation in place
Store Governance	Implemented	Create/open-close flows and merchant credential updates available
Driver Operations	Implemented	Driver CRUD plus credential management and order visibility
Audit Visibility	Implemented	Store and order logs exposed in admin reporting screens
Currency Handling	Implemented	KES display standardization available in UI formatting utilities

7. Current Risks and Improvement Priorities

High-Value Next Steps

1. Implement immutable audit trails with tamper-evident chaining for critical events.
2. Add formal export options (PDF/CSV) from admin logs for compliance workflows.
3. Expand driver analytics: on-time ratio, cancellation impact, SLA adherence.
4. Add account lifecycle controls for drivers (suspend/reactivate) and credential reset policy.
5. Introduce environment-specific deployment profiles and centralized secret management.

Medium-Term Enhancements

- Automated regression tests around role isolation and auth flows
- Advanced store-level performance dashboards (conversion, fulfillment latency)
- Notification workflow for key operational events (order risk, driver availability drop)

8. Conclusion

The platform is operationally structured with a clear role model and meaningful admin governance controls. Current implementation supports day-to-day store and fleet management while maintaining separation of access, credential controls, and event visibility. With targeted improvements in audit hardening, analytics depth, and reporting exports, Fikisha is well-positioned to scale in reliability and management maturity.

End of report. Generated from current workspace implementation snapshot on March 16, 2026.